

ECE275 Sample exam Spring 2026

Instructor: Vikas Dhiman (vikas.dhiman@maine.edu)

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Student Name:

Student Email:

1 Instructions

- There are seven problems. All problems are required.
- Maximum number of marks is 100. This exam amounts 10% toward the final grade.
- Time allowed is 120 minutes.
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8×11 in two-sided cheatsheet is allowed.
- Non-programmable calculators are permitted.
- Please use a pen or heavy pencil to ensure legibility. Colored pens/pencils are recommended for K-map grouping.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

Problem 1. *Simplify the following Boolean equation using Boolean theorems. Check for correctness using a K-map (5 marks).*

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}C\bar{D} + ABD + \bar{A}\bar{B}C\bar{D} + B\bar{C}D + \bar{A}$$

- Problem 2.** Implement the function $Y = BC + \bar{A}\bar{B}\bar{C} + B\bar{C}$ using
- (a) an 8:1 multiplexer (5 marks)
 - (b) a 4:1 multiplexer and no other gates (5 marks)
 - (c) a 2:1 multiplexer, one OR gate, and an inverter (5 marks)

Problem 3. Your company, Detect-o-rama, would like to design an Finite State Machine (FSM) that takes two inputs, A and B , and generates one output, Z . The output in cycle n , Z_n , is either the Boolean AND or OR of the corresponding input A_n and the previous input A_{n-1} , depending on the other input, B_n :

$$Z_n = A_n A_{n-1} \quad \text{if } B_n = 0 \quad (1)$$

$$Z_n = A_n + A_{n-1} \quad \text{if } B_n = 1 \quad (2)$$

(a) Is this FSM a Moore or a Mealy machine? (2 marks)

(b) Draw the state transition table (18 marks).

Use the following header for the state table

Present State A_{n-1}	Next State				Outputs (Z_n)			
	Inputs $A_n B_n =$				$00 \quad 01 \quad 10 \quad 11$			
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	1	1	1

Problem 4. A sequential circuit is to be used to control the operation of a vending machine which dispenses a \$0.25 product. The circuit has three inputs (N, D, and Q) and two outputs (R and C). The coin detector mechanism in the vending machine is synchronized with the same clock as the sequential circuit you are to design. The coin detector outputs a single 1 to the N, D, or Q input for every nickel (5 cents), dime (10 cents), or quarter (25 cents), respectively, that the customer inserts. Only one input will be 1 at a time. When the customer has inserted at least \$0.25 in any combination of nickels, dimes, and quarters, the vending machine must give change and dispense the product. The coin return mechanism gives change by returning nickels to the customer. For every 1 output on C, the coin return mechanism will return one nickel to the customer. The product is dispensed when the circuit outputs a single 1 on output R. The circuit should reset after dispensing the product.

Example: The customer inserts a nickel, a dime, and a quarter. The circuit inputs and outputs could look like this:

Inputs:

	0	5	10	25	40
N=	0	1	0	0	0
D=	0	0	1	0	0
Q=	0	0	0	1	0
R=	0	0	0	0	1
C=	0	0	0	0	0

N	D	Q
1	1	1
0	0	0
1	1	1

Note that any number of 0's can occur between 1 inputs. Complete the following Moore state table for the sequential circuit, and for each state indicate how much money the customer has inserted or how much change is due. Use the following header for the state table (PS means Present state)

Money inserted	Change due	PS	Inputs				Next State	Output (RC)
			NDQ = 000	NDQ = 100	NDQ = 010	NDQ = 001		
0	0	S ₀	S ₀	S ₁	S ₂	S ₃	00	
(30 marks)								
5	0	S ₁	S ₁	S ₂	S ₄	S ₅	00	
10	0	S ₂	S ₂	S ₄	S ₆	S ₇	00	
25	0	S ₃	S ₀	S ₁	S ₂	S ₃	10	
15	0	S ₄	S ₄	S ₆	S ₃	S ₈	00	
30	5	S ₅	S ₃	S ₅	S ₇	S ₈	01	
20	0	S ₆	S ₆	S ₃	S ₅	S ₉	00	
35	10	S ₇	S ₅	S ₇	S ₈	S ₉	01	
40	15	S ₈	S ₇	S ₈	S ₉	S ₉	01	
45	20	S ₉	S ₈	S ₉	S ₉	S ₉	01	

2. State reduction, 3. State encoding, 4. State encoded transition table

1. Truth table

Comb. circuit

Inputs				Outputs
A	B	C	D	γ
0	0	0	0	0

2. State transition table

Inputs		State		Next State		Output
X_2	X_1	Q_2	Q_1	Q_2^+	Q_1^+	$Z_2 Z_1$

3. Characteristic table : State transition table for a particular type of ff : D-ff / T-ff / J-K ff

4. Excitation table : State $\rightarrow$ Next State | Inputs

5. State encode transition table : State transition table with state encodings

↓ State encoding

States	encoding		
	Q_2	Q_1	Q_0
s_0	0	0	1
s_1	1	1	0

Problem 5. (30 marks) Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:

1. There is no direct connection from Rome to Trento
2. There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.
3. Regional trains connect towns in the same region
4. Intercity trains don't stop in small towns.

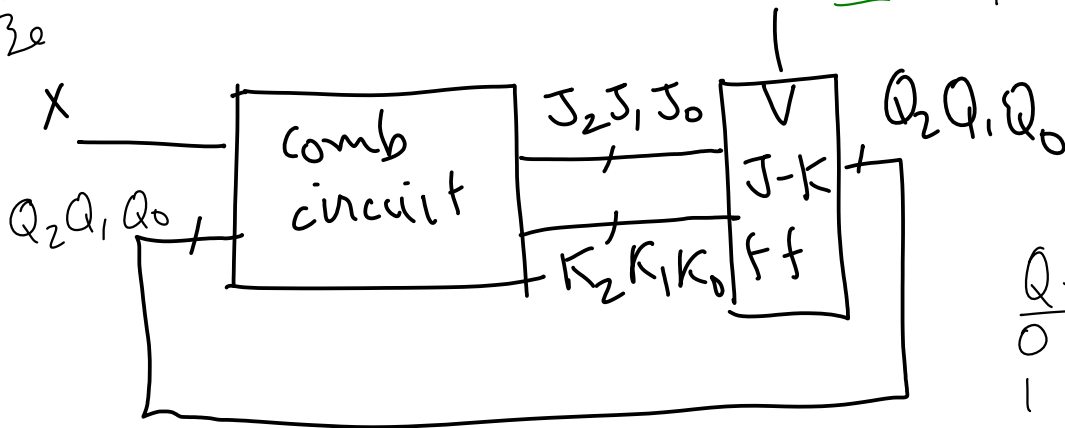
Problem 6. Find the expression for J_1 and K_1 assuming that J_1 and K_1 are inputs to the J-K flip flop that capture the state of the second most significant bit Q_1 of the following state encoded table. The state encoding table given with state encoding denoted as $Q_2Q_1Q_0$. Here X is the input and Z is the output. (20 marks).

Set J K		Q^+		Present State		Next State		Output	
				$Q_2Q_1Q_0$		$X=0$ $Q_2^+Q_1^+Q_0^+$	$X=1$ $Q_2^+Q_1^+Q_0^+$	$X=0$ Z	$X=1$ Z
0 0		Q	hold	000	0	100	8 101	1	0
0 1		0	Reset	001	1	100	9 101	0	1
1 0		1	set	010	2	000	10 000	1	0
1 1		$\bar{Q}$	toggle	011	3	000	11 000	0	1
				100	4	111	12 110	1	0
				101	5	110	13 110	0	1
				110	6	011	14 010	1	0
				111	7	011	15 011	0	1

Excitation
Table for J-K ff

$Q \rightarrow Q^+$	J	K	
0 0	0	d	Reset
0 1	0	d	hold
1 0	1	d	toggle
1 1	d	0	Set
	d	0	Set
			hold

Memorize



$Q \rightarrow Q^+$	J	K
0	Q^+	d
1	d	$\bar{Q}^+$

Q_1^+

$Q_1=0$	0	1	1	0
	0	1	1	0
$Q_1=1$	0	1	1	0
	0	1	1	0

$Q_2=0$ $Q_2=1$ $Q_2=0$

J_1
 $y \text{ } Q_1=0: J_1=Q_1^+$
 $y \text{ } Q_1=1: J_1=d$

J_1

0	1	1	0
0	1	1	0
d	d	d	d
d	d	d	d

Q_2

$J_1=Q_2$

K_1
 $y \text{ } Q_1=0: K_1=d$
 $y \text{ } Q_1=1: K_1=\bar{Q}_1^+$

K_1

d	d	d	d
d	d	d	d
1	0	0	1
1	0	0	1

$Q_2=0$ $Q_2=1$ $Q_2=0$

$K_1=\bar{Q}_2$